Bhutan Food & Drug Authority (BFDA)

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1. System Information

System Name : Bhutan Biosecurity Assessment System (BBAS)

Agency Name: Bhutan Food and Drug Authority

2. Introduction

The Bhutan Food and Drug Authority (BFDA) is mandated to enforce on-farm biosecurity requirements in livestock farms to prevent and control animal disease outbreaks. To support uniform implementation of this standard, BFDA developed a biosecurity checklist currently used during field assessments.

At present, assessments are conducted using a paper-based checklist (over 10 pages per farm), which results in high paper consumption and significant manual work. In FY 2024–2025, BFDA assessed 919 poultry and piggery farms, and this number is expected to increase with the expansion of commercial livestock farming. The current approach is time-consuming and requires substantial human and logistical resources for data collection, verification, compilation, storage, and reporting.

This BRD describes the requirements for a digital Bhutan Biosecurity Assessment System to streamline farm assessment, standardize assessments, enable offline operations, automate scoring/reporting, and integrate with the existing farm registration, Poultry and Piggery Farm Registration System (PPFRS).

3. Purpose

This Business Requirements Document (BRD) defines the business requirements for implementing a Biosecurity Assessment System for BFDA to:

- Conduct on-farm biosecurity assessments using a digital checklist.
- Capture assessment observations and evidence. - capturing pictorial/video by the app would enable a factual assessment.
- Automatically calculate scores and assign a compliance status.
- Maintain a history of assessments and non-compliances of each farm for follow-up.
- Exchange required farm and assessment outcome information with the Poultry and Piggery Farm Registration System (PPFRS).

4. Background and Rationale (Business Context)

4.1 Current Challenges

√ The checklist is long and paper-based, creating:

- Repetitive paperwork,
- Manual data entry,
- Bulky file storage,
- Time consuming, hence delays in compilation and reporting.
- Resource intensive and not environment friendly

√ Manual calculations increase the risk of errors and inconsistencies.

√ Data analysis and reporting are slow due to manual aggregation.

4.2 Rationale for Digitization

√ A digital Bhutan Biosecurity Assessment System will:

- Reduce administrative burden and costs associated with printing/photocopying/storage.
- Improve data quality through built-in validation and standardized formats.
- Enable automatic scoring and instant report generation will save time.
- Support offline data collection with synchronization when connectivity is available.
- Provide exportable datasets (Excel) for faster analysis, reporting, and decision-making.
- Record the exact location of farms visited (capture geocoordinates)
- No paper use directly contributes to environmental conservation policy
- Less time consumption
- Biosecurity assessment findings can be shared easily with the PPFRS so that the responsible officials in the Department of Livestock can technically support the farmers concerned

4.3 Related Systems and Ownership

√ PPFRS (Poultry and Piggery Farm Registration System) is used for registration of poultry and piggery farms and is owned/operated by the Department of Livestock (DoL) (external stakeholder).

√ BFDA is responsible for farm biosecurity assessment, including completing checklists and determining compliance.

- √ The Bhutan Biosecurity Assessment System must access farm registration data maintained in PPFRS and push assessment outcomes back to PPFRS through agreed integration mechanisms.

5. Definitions / Abbreviations

Term/Abbreviation	Definition
BFDA	Bhutan Food and Drug Authority
BBAS	Bhutan Biosecurity Assessment System
DoL	Department of Livestock
MoAL	Ministry of Agriculture and Livestock
PPFRS	Poultry and Piggery Farm Registration System (owned by DoL)
Bhutan Biosecurity Assessment System	BFDA's digital system for farm biosecurity assessments, scoring, and reporting
Biosecurity Assessment	Farm assessment carried out by BFDA field officials using the checklist
Checklist	Standard on-farm biosecurity assessment tool designed based on biosecurity standards issued by MoAL
Score	Computed result based on checklist responses (e.g., 0–3 per prompt)
Status	Compliance outcome derived from scoring and/or rules (e.g., Compliant/NC)
NC	Non-Compliance (failure to meet required biosecurity criteria those scoring 'fair' or 'poor' under each prompt)
System of Record (SoR)	Authoritative source system for a given data domain
Registered Farm	
Non-Registered Farm	

6. Objectives

6.1 Overall Objective

To develop a user-friendly, standalone digital system (compatible with smart mobile phones, tablets, and similar devices) for standardized on-farm biosecurity assessment and enforcement, with functionality to export collected data to Excel for analysis.

6.2 Specific Objectives

- √ Enable BFDA to perform standardized biosecurity assessment using digital checklists.
- √ Enable biosecurity assessments for:

- Registered farms (originating from PPFRS), and
- Non-registered farms encountered in the field.

√ Ensure consistent and transparent scoring and compliance status assignment.

√ Support controlled and auditable data exchange between PPFRS (DoL) and the Bhutan Biosecurity Assessment System (BFDA).

√ Improve operational efficiency through offline capability, automated reporting, and centralized assessment history.

7. Scope

7.1 In Scope

A) Farm Data Availability (PPFRS → BBAS)

√ Synchronization of farm data from PPFRS to BBAS for:

- New farm registrations
- One-time migration of existing registered farms (initial load)
- Farm deregistration updates
- Farm ownership change updates

B) Biosecurity Assessment in BBAS

√ Biosecurity Assessment process including:

- Conduct assessment for registered farms
- Conduct assessment for non-registered farms (by capturing minimum farm/farmer details)
- Capture checklist responses, observations, and evidence (as defined by BFDA)
- Automated scoring and compliance status determination
- Storage of non-compliance history for follow-up actions

C) Sharing Outcomes (BBAS → PPFRS)

√ Publishing biosecurity assessment outcomes to PPFRS:

- Push assessment score and compliance status back to PPFRS based on agreed trigger rules (minimum: when NC is found) both for new registration, scheduled or ad-hoc assessments.

D) Key System Features

√ Digital checklist with a user guide on scoring

√ Offline and online usability with synchronization

√ Automatic scoring and report generation

√ Exportable datasets in MS Excel

√ Color coding/visual prioritization based on overall score

√ Role-based access for field and administrative users

8. Stakeholders & Ownership

8.1 Key Stakeholders

- **BFDA:** Owner of Bhutan Biosecurity Assessment System and assessment processes
- **DoL:** Owner of PPFRS and farm registration processes
- **Biosecurity Inspectors / Field Officials:** Conduct assessments and enter data
- **BFDA Project Team:** Implements Bhutan Biosecurity Assessment System and manages operations/support

8.2 System of Record (SoR)

- **PPFRS (DoL)** is SoR for:
  - Farm registration details
  - Deregistration
  - Ownership records/changes
- **Bhutan Biosecurity Assessment System (BFDA)** is SoR for:
  - Assessment events
  - Checklist responses and evidence
  - Scoring and assessment compliance outcomes

9. Current State (As-Is)

1. Farms are registered and maintained in PPFRS.
2. Biosecurity assessment are conducted using paper checklists with heavy manual processes.
3. Assessment outcomes are not systematically available in a structured, automated way for analysis and reporting.
4. There is no consistent system-to-system mechanism to push assessment outcomes back to PPFRS.

10. Future State (To-Be)

1. When a farm is registered/updated in PPFRS, the record becomes available in the Bhutan Biosecurity Assessment System for assessment planning and execution.
2. BFDA officials conduct farm biosecurity assessments using the Bhutan Biosecurity Assessment System checklist.
3. The Bhutan Biosecurity Assessment System calculates a score and assigns a compliance status.

4. The Bhutan Biosecurity Assessment System pushes assessment outcomes (score and status) back to PPFRS through an agreed API mechanism.
5. Bhutan Biosecurity Assessment System can also record assessment outcomes for farms not found in PPFRS (non-registered farms), as per BFDA operational needs.

11. Desired System / Solution

11.1 System Overview

BBAS will be implemented as a **mobile-first Progressive Web Application (PWA)** with an offline-first architecture, supported by a centralized web-based supervisor portal.

The mobile-first design ensures usability on smartphones and tablets used by field officials. The offline-first architecture enables inspectors to complete full assessments without internet connectivity, with secure local storage and synchronization when connectivity becomes available.

The system will replace the paper checklist with a standardized digital format and provide structured data capture, automated scoring, and centralized reporting.

11.2 Core Capabilities

1. Offline Inspection

- Complete checklist capture without internet.
- Local storage of responses, comments, GPS coordinates, and photos.
- Auto-save and draft functionality.

2. Digital Evidence Capture

- In-app photo capture with compression.
- GPS tagging of assessments.
- Controlled number of photos per checklist item.

3. Secure Synchronization

- Manual and automatic sync when connectivity is available.
- Resumable uploads to prevent data loss.
- Clear sync status indicators (Draft, Queued, Synced, Failed).

4. Hybrid Authentication

- Preferred: Login before inspection; assessments directly linked to authenticated user.
- Fallback: Provisional offline capture allowed when not logged in.
- Mandatory authentication during sync for provisional records.
- Complete audit trail linking device ID, timestamps, and user identity.

5. Supervisor Portal

- View and filter submitted assessments.

- Access photos and GPS coordinate information.
- Review provisional or flagged records.
- Basic compliance and non-conformance reporting.

11.3 Security and Governance

- Secure HTTPS communication.
- Token-based authentication.
- Role-based access control.
- Audit logging for traceability.
- Local data protection for sensitive information.

11.4 Non-Functional Priorities

- Optimized for low- and mid-range Android and commonly used iOS devices.
- Capable of handling multiple offline inspections with photos.
- Reliable sync recovery after network interruption.
- Maintainable single web-based codebase.

11.5 Pilot and Rollout

The system will be piloted in selected dzongkhags to validate offline performance, media handling, synchronization reliability, and supervisor workflows before national rollout.

12. System Integration (PPFRS ↔ BBAS)

12.1 Integration Principles

√ PPFRS is an external system owned by DoL; integration must be governed by inter-agency agreements.

√ Data exchange must be secure, auditable, and version-controlled.

√ The Bhutan Biosecurity Assessment System must not override PPFRS-owned master data fields except as explicitly agreed.

12.2 Integration Interfaces (High-Level)

A) PPFRS → BBAS (Inbound)

√ Farm registration create/update (new registrations)

√ Deregistration events

√ Ownership change events

√ One-time bulk extract/import for existing farms

B) BBAS → PPFRS (Outbound)

√ Assessment outcome notification:

- PPFRS Farm ID

- Assessment ID (BBAS)
- Assessment date
- Score
- Compliance status
- NC (multiple records)

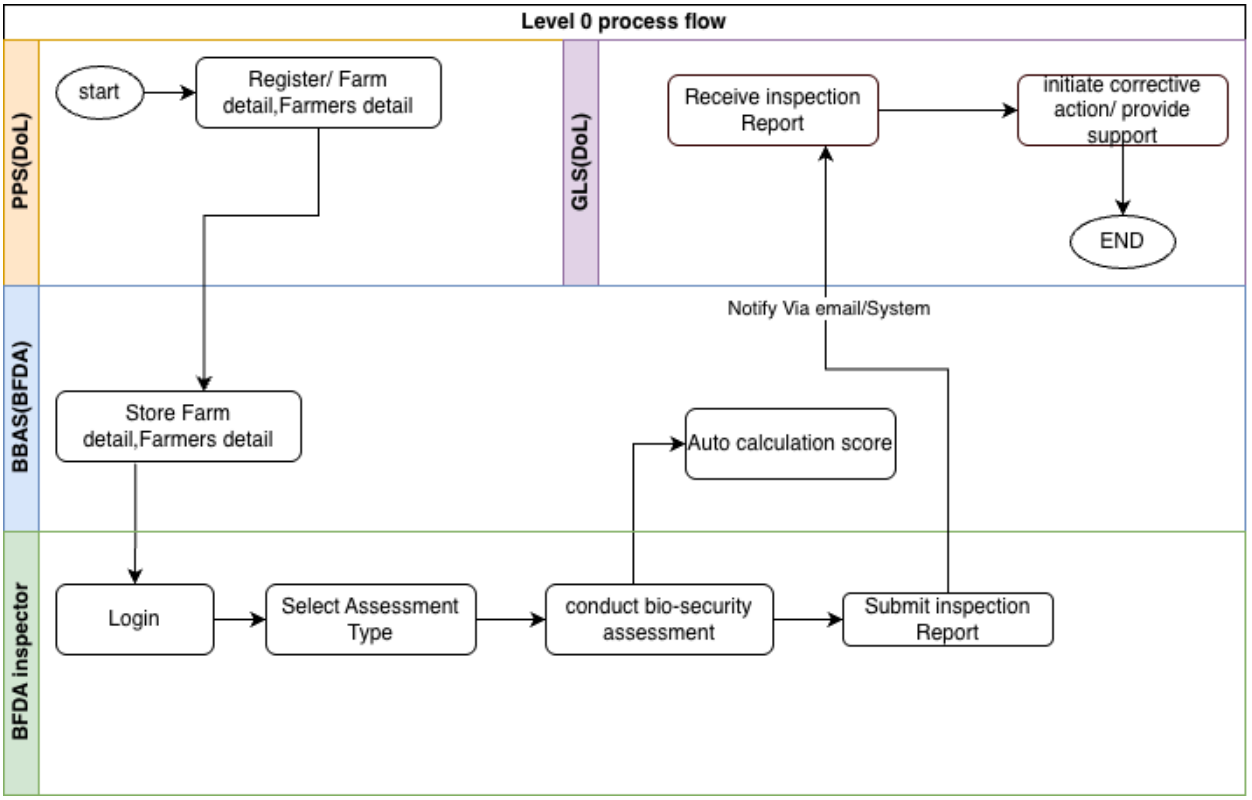
12.3 Trigger Rule (Minimum Requirement)

√ When Non-Compliance (NC) is identified during an assessment, the BBAS shall push the score, status, and NCs to PPFRS.

13. Business Process & Integration Overview

13.1 Farm Registration Sync (PPFRS → BBAS)

Process



1. A farm is registered in PPFRS.
2. PPFRS pushes the farm details to the BBAS via **Push API**.
3. Bhutan Biosecurity Assessment System stores/updates the farm record.
4. Biosecurity assessment checklist scoring/non-compliance push back to PPFRS via API

Business Requirement

- Bhutan Biosecurity Assessment System must accept farm master data from PPFRS and maintain it for assessment selection.

13.2 One-Time Migration of Existing Farms (PPFRS → BBAS)

Process

1. DoL provides a one-time export of existing farm registrations from PPFRS.
2. Bhutan Biosecurity Assessment System imports the dataset and validates duplicates using PPFRS identifiers.

Business Requirement

- Bhutan Biosecurity Assessment System must support initial bulk import/migration and reconciliation using unique farm identifiers.

13.3 Biosecurity Assessment — Registered Farms

Process

1. BFDA Officials selects a registered farm (synced from PPFRS).
2. The BFDA officials completes the assessment .
3. Bhutan Biosecurity Assessment System calculates **score** and determines **Non-Compliance**.
4. Bhutan Biosecurity Assessment System pushes score and non compliance back to PPFRS (per trigger rules below).

Trigger Rule (from current requirement)

- If **Non-Compliance (NC)** is found, the system must trigger and push the **scoring and NC** to PPFRS.

13.4 Biosecurity Assessment - Non-Registered Farms

Process

1. The BFDA officials select the "Non-Registered Farm" option.
2. BFDA officials enter minimum **Farm and Farmer details** (as defined by BFDA).
3. The BFDA officials complete the assessment and record results.

Business Requirement

- The Bhutan Biosecurity Assessment System must allow capturing assessment records for non-registered farms, including required identifiers and location details.

13.5 Deregistration Updates (PPFRS → BBAS)

Process

1. The farm is deregistered in PPFRS.
2. PPFRS pushes deregistration update to BBAS.
3. Bhutan Biosecurity Assessment System updates farm status to prevent inappropriate assessment scheduling and to preserve historical assessment records.

13.6 Ownership Change Updates (PPFRS → BBAS)

Process

1. Ownership changes in PPFRS.
2. PPFRS pushes ownership change update to BBAS.
3. Bhutan Biosecurity Assessment System updates the farm's ownership details while maintaining assessment history.

13.7 Follow-Up Workflow (Operational Logic)

1. When an assessment identifies one or more Non-Compliances (NC), the system shall automatically mark the assessment as "NC – Follow-Up Required."

2. The system shall generate a follow-up task linked to the original assessment ID.
3. Follow-up assessment shall not be permitted before three (3) calendar days from the initial assessment date.
4. The assigned Focal Officer shall receive notification of pending follow-up cases.
5. NC items shall remain in “Active Non-Compliance” status until verified compliant during follow-up assessment.
6. Once compliance is verified, the NC shall be marked “Closed” and removed from the active NC list but retained in historical records for audit purposes.
7. If follow-up assessment again identifies NC, the cycle shall continue until compliance is achieved.

14. Biosecurity check list

Key characteristics:

- Covers **end-to-end farm biosecurity practices**
- Structured into 5 domains:
  - Farm premises
  - Farm shed facilities
  - Herd health practices
  - Feed & equipment
  - Record keeping
- Uses **scoring mechanism (0–3)** for each prompt
- Produces **overall biosecurity score** (Poor → Excellent)

Farm Biosecurity Assessment Checklist:

1.1 Farm Details

Field Name	Input Type	Validation	Remarks
Type	Radio button	Required	Options: **Registered Farm** / **Not-Registered Farm**. Toggles form behavior accordingly.
Registered Farm			
Farm Id	Textbox	Required	On change of Farm Id, fetch following details from Farm details table.
Registration Date	Display	-	-
Farm Type	Display	-	-
Farm Name	Display	-	-
Farmer CID	Display	-	-
Farmer Name	Display	-	-
Dzongkhag (Farm)	Display	-	-
Gewog (Farm)	Display	-	-
Village (Farm)	Display	-	-
Mobile No	Display	-	-
Not Registered Farm			
Farmer CID	Textbox	Required, Numeric 11 digits (CID). Text for Permit	On change of CID Number, the system should call the API and pull Name. Manual entry for Permit holders.
Farmer Name	Textbox	Required	Fetch Name from citizen API.

			Manual entry for Non-CID
Dzongkhag (Farm)	Dropdown	Required	Fetch from master Dzongkhag
Gewog (Farm)	Dropdown	Required	Fetch from master Gewog
Village (Farm)	Dropdown	Required	Fetch from master Village
Mobile No	Textbox	Required, Numeric, 8 digits	Bhutan phone number format

1.2 Farm Premise Facility & Practices

Field Name	Input Type	Validation	Lookup Table
Have a biosecurity signboard at the entrance of the farm ?-> [cautioning visitors to follow biosecurity procedures, contains information on farm name, name of owner/manager and his contact number]	Select	Pull from Master Table	0 for Poor 1 for Fair 2 for Good 3 for Excellent
Farm has a functional perimeter fence	Select	Pull from Master Table	
Have functional foot dip at the entrance gate of farm	Select	Pull from Master Table	
Have functional tyre dip at the entrance of the farm: ? -> If vehicles can cross the entrance gate of perimeter fencing and enter the farm, the tyre dip is mandatory.	Select	Pull from Master Table	
Have a wash station (tap water, scrubbing brush, detergent) at the entrance of the farm	Select	Pull from Master Table	
Have farm gears for workers (boots/shoes and farm clothes)	Select	Pull from Master Table	
Farm distance from the highway	Select	Pull from Master Table	
Farm Premise Facility & Practices [Biosecurity Score]	Auto [total of all divided by number of questions]		

Auto-sum all the assessed fields, and divide by the number of assessed fields. Same applies to all the five domains.

1.3 Farm Shed Facility & Practices

Field Name	Input Type	Validation	Lookup Table

The shed surroundings are clean	Select	Pull from Master Table	0 for Poor 1 for Fair 2 for Good 3 for Excellent
Have good drainage system in place	Select	Pull from Master Table	
Have functional foot dip at the entrance of each shed	Select	Pull from Master Table	
Have separate shed for isolation of sick animals	Select	Pull from Master Table	
Have a separate shed for quarantine (new arrival)	Select	Pull from Master Table	
Animal shed is kept clean (free of physical/extraneous matters) and disinfected	Select	Pull from Master Table	
Farm Shed Facility & Practices [Biosecurity Score]	Auto [total of all divided by number of questions]		

1.4 Herd Health Practices

Field Name	Input Type	Validation	Lookup Table
Vaccination as per schedule (refer record for one year at least)	Select	Pull from Master Table	0 for Poor 1 for Fair 2 for Good 3 for Excellent
Deworming at regular interval (refer record for one year at least)	Select	Pull from Master Table	
Check the animals/birds for any signs of disease at least twice a day (refer record for validation)	Select	Pull from Master Table	
Isolate and treat sick animals/birds (refer treatment record)	Select	Pull from Master Table	
Report any sick or deaths of animal/bird to nearest animal health center or BFDA office within 24 hours.	Select	Pull from Master Table	
Type of feeding practices followed	Select	Pull from Master Table	
Source of water supply to the farm	Select	Pull from Master Table	
Disposal of dead animal carcass (visit the site and see)	Select	Pull from Master Table	
Restrict visitors or brief essential visitors on the biosecurity measures of the farm (refer visitor record)	Select	Pull from Master Table	

Herd Health Practices [Biosecurity Score]	Auto [total of all divided by number of questions]		

1.5 Feed and Equipment Facility

Field Name	Input Type	Validation	Lookup Table
Feed and feed storage	Select	Pull from Master Table	0 for Poor 1 for Fair 2 for Good 3 for Excellent
Farm equipment use and maintenance. (Check the farm equipment)	Select	Pull from Master Table	
Feed and Equipment Facility [Biosecurity Score]	Auto [total of all divided by number of questions]		

1.6 Other Essential Record Keepings

Field Name	Input Type	Validation	Lookup Table
Morbidity and mortality record maintained	Select	Pull from Master Table	0 for Poor 1 for Fair 2 for Good 3 for Excellent
Animal stock record maintained	Select	Pull from Master Table	
Visitor's record maintained with [Name of person, company affiliation, date of visit, phone number, reason for visit, date and description of recent visits to other farms, if applicable]	Select	Pull from Master Table	
Other Essential Record Keepings [Biosecurity Score]	Auto [total of all divided by number of questions]		

1.7 Overall biosecurity score

Field Name	Input Type	Validation	Lookup Table
Overall Biosecurity score	Auto [Overall biosecurity score = Sum of totals of 1.1 to 1.5 divided by 5		0 for Poor 1 for Fair

			2 for Good 3 for Excellent

√ IF Non-Compliance Found in any of the check-list, or all Non-compliance , it should push the NC details to PPFRS. All the fields with a score of Fair (score of 1) or Poor (score of 0) shall be listed as Non-compliant with the corresponding description of the score pulled from the scording guide.

√ The NC list must be recorded separately.

√ The follow up button must be enabled only in 3 days and the Focal must be notified and must do the followup again.

√ It will be in loop unless compliance. Once the compliance result is shown, the list must be removed from the non-compliance list.

15. Functional Requirements (FR)

15.1 Farm Master Data

√ **FR-01**: BBAS shall receive farm registration data from PPFRS through secure API integration.

√ **FR-02**: BBAS shall store and display farm master records for assessment selection.

√ **FR-03**: BBAS shall support one-time bulk migration of existing registered farms from PPFRS during initial setup.

√ **FR-04**: BBAS shall update farm status upon receiving deregistration events from PPFRS and prevent new assessments for deregistered farms while retaining historical records.

√ **FR-05**: BBAS shall update farm ownership details upon receiving ownership change events from PPFRS while preserving assessment history.

15.2 Assessment Creation and Management

√ **FR-06**: BBAS shall allow authorized inspectors to create assessments for registered farms.

√ **FR-07**: BBAS shall allow inspectors to create assessments for non-registered farms by capturing minimum farm and farmer details as defined by BFDA.

√ **FR-08**: During assessment, BBAS shall capture the following farm assessment overview details:

- Farm type (e.g., poultry, piggery)
- Number of animals/birds present at the time of assessment
- Farm category (commercial, semi-commercial, subsistence)
- Farm location (Dzongkhag, Gewog, GPS coordinates)
- Contact details of farm owner

√ **FR-09**: BBAS shall capture GPS coordinates automatically during assessment.

√ **FR-10**: BBAS shall support offline creation, saving, and editing of assessments.

√ **FR-11**: BBAS shall auto-save assessment progress locally when operating offline.

√ **FR-12**: BBAS shall allow synchronization of offline assessments once internet connectivity is restored.

15.3 Checklist & Scoring Logic

√ **FR-13**: BBAS shall render the digital biosecurity checklist as approved by BFDA, structured by defined domains.

√ **FR-14**: The checklist shall evaluate the following biosecurity measures for each farm:

- Availability of fencing
- Availability of biosecurity signboard
- Presence of footbath at farm entrance and shed entry
- Presence of vehicle tyre dip at the entrance of the farm
- Availability of wash station at the entrance of the farm
- Availability of designated farm gears
- Distance of farm from highways
- Cleanliness of shed surroundings
- Cleanliness of animal shed
- Availability of proper drainage system in the farm
- Availability of visitor logbook
- Availability of isolation/quarantine facilities
- Daily monitoring of animal health
- Carcass and waste management practices
- Implementation of pest control measures
- Vaccination and deworming status
- Practice of reporting sickness/deaths to authorities
- Feeding practices & feed storage conditions
- Water source
- Treatment records
- Morbidity and mortality records
- Animal stock records
- Farm equipment use and maintenance

√ **FR-15**: BBAS shall restrict checklist scoring inputs to predefined values (0–3) as per scoring standards.

√ **FR-16**: BBAS shall automatically calculate domain-level biosecurity scores.

√ **FR-17**: BBAS shall automatically calculate overall biosecurity score based on defined formula.

√ **FR-18**: BBAS shall determine compliance status based on scoring thresholds defined by BFDA.

√ **FR-19**: BBAS shall automatically classify farm biosecurity performance into the following categories based on score:

- Poor
- Fair
- Good
- Excellent

√ **FR-20**: BBAS shall categorize farms into risk levels (High-Risk, Medium-Risk, Low-Risk) based on configurable scoring thresholds.

√ **FR-21**: BBAS shall automatically identify and flag Non-Compliance (NC) items.

15.4 Non-Compliance & Follow-Up Management

√ **FR-22**: BBAS shall record all Non-Compliance (NC) items separately from compliant responses.

√ **FR-23**: BBAS shall automatically generate a follow-up requirement when NC is identified.

√ **FR-24**: BBAS shall restrict follow-up assessment initiation until at least three (3) calendar days after the initial assessment.

√ **FR-25**: BBAS shall notify the assigned Focal Officer or concerned livestock office when a follow-up is due.

√ **FR-26**: BBAS shall maintain NC records in active status until marked compliant through follow-up assessment.

√ **FR-27**: Upon compliance verification, BBAS shall close the NC while retaining historical records for audit and reporting purposes.

15.5 Comparative Analysis & Compliance Monitoring

√ **FR-28**: BBAS shall enable comparative analysis of biosecurity scores based on:

- Farm type (e.g., poultry, piggery)
- Number of animals/birds
- Farm category (commercial, semi-commercial, subsistence)
- Dzongkhag

√ **FR-29**: BBAS shall classify locality-level biosecurity compliance as:

- High compliance (≥80% of assessed fields scores Good or Excellent)
- Moderate compliance (60–79% of assessed fields scores Good or Excellent)
- Low compliance (<60% of assessed fields scores Good or Excellent)

√ **FR-30**: BBAS shall provide dashboards and summary views for monitoring compliance trends at Dzongkhag and national levels.

15.6 Reporting & Information Sharing

√ **FR-31**: BBAS shall generate comprehensive biosecurity assessment reports including:

- Total number of registered farms
- Total number of assessed farms
- Number of compliant farms
- Number of non-compliant farms
- Best-performing Dzongkhags

√ **FR-32**: BBAS shall support export of reports in standard formats (e.g., Excel).

√ **FR-33**: BBAS shall support secure API-based information sharing with the Department of Livestock (DoL).

√ **FR-34**: BBAS shall support risk communication mechanisms to notify concerned livestock offices regarding high-risk farms.

15.7 Outcome Sharing to PPFRS

- √ **FR-35**: BBAS shall push assessment ID, assessment date, score, compliance status, and NC indicators to PPFRS via API when NC is identified (minimum requirement).
- √ **FR-36**: BBAS may support pushing outcomes for all completed assessments subject to inter-agency agreement.

16. Data Requirements (High-Level)

The following minimum data elements shall be exchanged and stored within BBAS. The final data dictionary shall be jointly agreed between BFDA and DoL during detailed design.

16.1 Data Exchange: PPFRS → BBAS

The following data elements shall be received from the PPFRS system:

Farm Identification Data

- PPFRS Farm ID (unique identifier)
- Farm name
- Farm type (e.g., poultry, piggery)
- Farm category (commercial, semi-commercial, subsistence)
- Dzongkhag
- Gewog
- Village (if available)
- Registration status (active/deregistered)
- Date of registration
- Deregistration indicator (if applicable)

Owner Information

- Owner name
- Owner identifier (as permitted)
- Contact number
- Ownership change details (if applicable)

Farm Profile Data

- Total number of animals/birds (as recorded in PPFRS)

16.2 Data Captured by BBAS During Assessment

The system shall capture and store the following assessment-related data:

Assessment Metadata

- Assessment ID (BBAS unique identifier)
- Linked PPFRS Farm ID (if registered)
- Assessment date and time
- Inspector user ID and Name, his/her office name
- Device ID (for audit purposes)
- GPS coordinates at time of assessment
- Checklist version number

Farm Assessment Overview

- Farm type (if editable at time of assessment)

- Number of animals/birds present at assessment
- Farm category (if applicable)
- Farm location details
- Contact details of farm owner (for non-registered farms)

Checklist & Scoring Data

- Checklist item responses
- Individual item scores (0–3 scale)
- Domain-level scores
- Overall biosecurity score
- Compliance status (Compliant / Non-Compliant)
- Performance category (Poor, Fair, Good, Excellent)
- Risk category (High, Medium, Low)

Non-Compliance Data

- List of identified NC items [all those fields with a score of Poor or Fair]
- NC status (Active / Closed)
- Date of NC identification
- Follow-up assessment reference ID (if applicable)
- Date of compliance verification

Evidence Data

- Photo file references
- Photo metadata (timestamp, GPS coordinates if available)

16.3 Data Exchange: BBAS → PPFRS

The following minimum assessment outcome data shall be shared with PPFRS:

- PPFRS Farm ID
- Assessment ID (BBAS)
- Assessment date/time
- Checklist version
- Overall score
- Compliance status
- Non-Compliance indicator [fields with the scores of Poor and Fair]
- Risk category (if agreed during integration design)

Data exchange shall occur via secure API using authenticated communication protocols.

16.4 Reporting & Aggregated Data Elements

For dashboard and reporting purposes, the system shall support generation of aggregated data including:

- Total registered farms
- Total assessed farms
- Total compliant farms
- Total non-compliant farms
- Compliance percentage by Dzongkhag
- Risk distribution (High/Medium/Low)
- Performance distribution (Poor/Fair/Good/Excellent)

16.5 Data Governance & Retention

- All assessment records shall be retained for audit and regulatory purposes.
- Historical assessment data shall not be deleted upon farm deregistration.
- All shared data shall comply with applicable data protection and inter-agency data sharing policies.
- Risk classification thresholds shall be configurable without altering historical records.

17. System Administration & User Management

17.1 User Roles

The Bhutan Biosecurity Assessment System shall support role-based access control. The following user roles are defined:

1. Biosecurity Inspector (Field Official)

- Conduct assessments (registered and non-registered farms)
- View checklist fields with assessor's guide for scoring
- Capture checklist responses and evidence
- View own assessment history
- Sync assessments (offline to online)
- Initiate follow-up assessments

2. BFDA Supervisor / Focal Officer

- View all assessments within assigned jurisdiction
- Review Non-Compliance (NC) cases
- Approve/validate follow-up assessments
- Monitor compliance status trends
- Access dashboards and reports
- Track unsynced or pending assessments

3. System Administrator (BFDA IT/Admin)

- Create, modify, deactivate user accounts
- Assign roles and access rights
- Manage checklist versions
- Manage master lookup tables
- Configure scoring rules (if required)
- Monitor system logs and audit trails
- Manage integration configuration with PPFRS

4. Read-Only / Management User (Optional)

- View reports and dashboards
- No data entry or editing permissions
- Download and export data in Excel for further analysis

17.2 User Authentication

- The system shall require secure login using username/password.
- Session management shall ensure automatic logout after inactivity.
- Password policies shall follow government ICT standards.

17.3 Role-Based Access Control (RBAC)

- Users shall only access data relevant to their assigned role and jurisdiction.
- Field officials shall not modify scoring logic.
- Supervisors shall not modify checklist schema unless explicitly authorized.
- Administrative functions shall be restricted to System Administrators.

17.4 Audit Trail

The system shall log:

- User login and logout events
- Assessment creation and modification
- Score calculations
- Follow-up actions
- Data synchronization events
- Data pushed to PPFRS

Audit logs shall not be editable by end users.

17.5 User Account Lifecycle Management

The system shall support:

- User creation
- Role assignment
- Temporary suspension
- Permanent deactivation
- Password reset
- Transfer of user jurisdiction (if official is reassigned)

18. Non-Functional Requirements

18.1 Performance

- System shall operate smoothly on mid-range Android devices.
- Assessment screens shall load within acceptable time under low connectivity.

18.2 Offline Capability

- Inspectors shall be able to complete full assessments without internet.
- Data shall be stored locally and synced when connectivity is restored.

18.3 Security

- All data exchange between BBAS and PPFRS shall use secure APIs (HTTPS).
- Role-based access control shall be enforced.
- Sensitive data shall not be accessible to unauthorized users.

18.4 Availability

- Server system shall maintain high availability during working hours.
- Backup and recovery mechanisms shall be implemented.

18.5 Data Retention

- Assessment records shall be retained as per BFDA policy.
- Historical data shall remain accessible for audit and reporting.

19. Dependencies

Dependencies

1. BFDA and DoL shall agree to share a minimal amount of data to support the performance of each other's system and the functions of the organizations.
2. Availability of PPFRS technical resources for API specifications, testing, and deployment coordination.
3. Definition and approval of the biosecurity checklist structure, scoring model, and compliance statuses.

20. Reports

The following list outlines the identified reports to be generated. Please note that the final set of reports may be refined during the design and development phases.

1. Biosecurity inspection Graph (Poultry, and Piggery) Gewog-wise, Dzongkhag wise
2. Disease outbreak alert
3. Risk based inspection (Automatic risk identification)
4. Biosecurity clearance report
5. **Compliance status**
6. **Detailed Non-Conformity (NC) Report-system can auto-flag overdue NCs.**
7. **Risk Categorization Dashboard Report**
8. Year-wise compliance trend
9. Dzongkhag comparison report
10. Farm performance improvement graph
11. Automatic notification to DLO, EC, BFDA

21. Business Requirements Document (BRD) Sign-Off

Date: [Date]

Project Name: [Project Name]

Document Overview

Provide a brief summary of the BRD, including the purpose and key objectives of the project. This section should include any critical assumptions or constraints highlighted in the document.

Attached BRD

Stakeholder Review and Approval

Please review the Business Requirements Document and provide your approval by signing below. Your signature confirms that you agree with the documented requirements and authorize the project to proceed based on these specifications.

Stakeholder Name	Role	Signature	Date	Comments (if any)
[Name]	[Role]	[Signature]	[MM/DD/YYYY]	[Comments or feedback]
[Name]	[Role]	[Signature]	[MM/DD/YYYY]	[Comments or feedback]
[Name]	[Role]	[Signature]	[MM/DD/YYYY]	[Comments or feedback]

Additional Notes

- Any additional comments or considerations can be noted here.
- Include a section for addressing any unresolved issues or action items.

Notes:

[Additional notes or comments]

Final Approval

The undersigned hereby approve the Business Requirements Document and agree to proceed with the project as specified.

Name	Role	Signature	Date
Mr.	Bhutan Food and Drug Authority		

This format ensures that all relevant stakeholders have the opportunity to review and approve the document, making it clear that the project can proceed based on the agreed-upon requirements. Adjust the template to fit the specific needs and processes of your organization.

22. Annexure: Biosecurity Assessment Checklist

Assessment Type:

- √ On Request / New Farm Registration
- √ Regular / Planned Assessment
- √ Follow Up: ***link to previous assessment***
- √ Complaint Based Assessment:
- √ Assessment During Disease Outbreak